Daniel Alejandro Silva Rojas

SENIOR FULL STACK SOFTWARE ENGINEER · NODE.JS · TYPESCRIPT · POSTGRESQL

Caracas, Venezuela · Remote (LATAM / US & Canada time zones) | +58 414-231-7561 | dsrglrm@gmail.com
danielsilvarojas.dev | github.com/Dan178A | linkedin.com/in/daniel-alejandro-silva-rojas

PROFESSIONAL SUMMARY

Full Stack Software Engineer with 5+ years of experience shipping and operating production software end to end: Node.js/TypeScript APIs, PostgreSQL data models, and modern SSR frontends. Founder and architect of dropaudioccs.com, a live e-commerce platform (Nuxt 3 + Supabase/PostgreSQL, multi-currency checkout, 110+ verified customer reviews) that I run in production. Until September 2026 I was Lead Performance & Systems Engineer at Ea2technology (Canada), where low-latency and concurrency work was my daily job. That systems background is what I bring to product teams: services that stay correct and fast under load, not just features that pass a demo. B.Sc. in Computer Science (LUZ-IUTA) completed in parallel with continuous professional work since 2021. English B2 (professional), with five years of daily work alongside a Canada-based team.

TECHNICAL SKILLS

Backend: Node.js (Express, NestJS, Nitro), TypeScript, REST API design, WebSockets (full-duplex), gRPC, TCP sockets, FastAPI / Python, Rust (Axum, Rocket)

Frontend: Vue 3, Nuxt 3 (SSR/Nitro), TypeScript, React / React Native, Tailwind CSS, Bootstrap 5, PWA

Databases: PostgreSQL (schema design, complex SQL, indexing, transactions, query optimization, RLS), MySQL, Redis, MongoDB, Supabase (Auth, Storage, Realtime, RLS)

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda), Docker, CI/CD with GitLab CI, GitHub Actions and AWS CodePipeline (CodeCommit, CodeArtifact, CodeBuild, CodeDeploy), Vercel

AI: RAG systems in Python, AI API integration (Gemini Live), AI-assisted development workflows

Also: Technical SEO/GEO, conversion-oriented UX, UI/UX design in Figma, low-latency architecture, concurrency, profiling

PROFESSIONAL EXPERIENCE

Founder & Software Architect

Jun 2021 – Present · part-time personal project

DropAudio CCS — dropaudioccs.com · e-commerce platform in production

- Own the full stack of a live revenue-generating store: Node.js server routes and API endpoints (Nuxt 3 / Nitro runtime), PostgreSQL data model, admin panel, catalog and checkout — 110+ verified customer reviews to date.
- Built a custom multi-currency checkout (USDT, Zinli, Pago Móvil) with live BCV/USDT exchange rates, order state handling and reconciliation, plus automated Web Push notifications driven by scheduled jobs (pg_cron).
- Designed the PostgreSQL schema with Row Level Security policies, indexes and optimized queries; every product, order and review path is authorization-checked at the database level.
- Developed a 3-step interactive recommender and technical comparison tool that reduced purchase friction and increased conversion.
- Engineered for continuity and discovery: fallback catalog for upstream outages, technical SEO/GEO (JSON-LD, sitemap) and PWA delivery, deployed on Vercel.

Lead Performance & Systems Engineer

Jul 2021 – Sep 2026 · full-time

Ea2technology — Canada (remote)

- Designed a low-latency, full-duplex WebSocket architecture for live operator dashboards, replacing HTTP polling with event-driven push: response latency dropped from 200–500 ms to 1–10 ms and bandwidth usage fell by up to 95%.
- Built and shipped an industrial Digital Twin platform (real-time telemetry for coke drum deformation in a refinery), taking high-frequency sensor data from ingestion through API layer to the interface operators use to prevent critical structural failures.
- Led the migration of legacy monolithic systems to cloud-native microservices on AWS, defining service boundaries, deployment flow and the trade-offs behind them.
- Optimized compute-intensive data pipelines in C++ and Rust for memory safety and concurrency under sustained load — the same discipline I apply to Node.js services under production traffic.
- Led end-to-end technical UI/UX design in Figma, and worked daily in English with a distributed team based in Canada.

Software Performance Consultant & Algorithm Engineer

Jan 2023 – Jul 2025

Freelance / Independent Consulting · Remote — client: Vessel Inspection and Assessment Corp (VIAAC), maritime/oil sector · role: Senior Developer

- Built and containerized backend services with NestJS and Docker, and set up the delivery pipelines around them: GitLab CI, GitHub Actions, and AWS CodePipeline with CodeCommit, CodeArtifact, CodeBuild and CodeDeploy.
- Ran 4 technology-stack audits, finding 3 memory leaks and concurrency bottlenecks in the processing of point clouds up to 1.2M points; the resulting optimizations cut a heavy sequential process from 12–14 hours to 5 hours and data-navigation response time from ~1 s to 15–20 ms.
- Presented audit findings and remediation standards to the client's development team, and specified the changes they carried into their own codebase.
- Delivered high-performance video stabilization engines (optical flow, matrix interpolation; Python, NumPy, SciPy, OpenCV) reaching 65–80% jitter reduction at 60 fps from HD to Full HD.
- Designed TCP socket and WebSocket communication systems with custom file-transfer protocols for high-frequency environments.

IT Support Specialist Jan 2021 – Sep 2021

RenéDessés de Venezuela — Caracas, Venezuela

- Sole IT/systems administrator for an 80-person company: Cisco network infrastructure, MySQL Server database, Profit business management system and Apache/Linux servers, plus day-to-day support for all 80 users.

FEATURED PROJECTS — CODE AT GITHUB.COM/DAN178A

DropAudio CCS · **Nuxt 3 / Node.js · PostgreSQL · Supabase · Vercel**
Production e-commerce platform with admin panel, catalog, multi-currency checkout and an interactive audio recommender.

extract_rif · **FastAPI · PaddleOCR · Docker**
HTTP microservice that extracts structured data from RIF tax documents (SENIAT) out of PDFs or images via a dual-engine OCR pipeline.

RealtimeVoiceAssistant · **Rust (Tokio) · Python · Gemini Live**
Real-time conversational assistant with streaming transcription, NLP and voice synthesis on an async Rust runtime.

EDUCATION

B.Sc. in Computer Science — LUZ-IUTA (Universidad del Zulia · University Institute of Technology), Venezuela · graduated April 2025 · completed while working full-time

Associate Degree (TSU) in Information Technology — LUZ-IUTA, Venezuela · 2018 – 2021

LANGUAGES & CERTIFICATIONS

Spanish — native · **English** — B2 professional (technical reading/writing, interviews, five years of daily work with a Canada-based team)

Introduction to Generative AI — Duke University (Coursera, 2025)